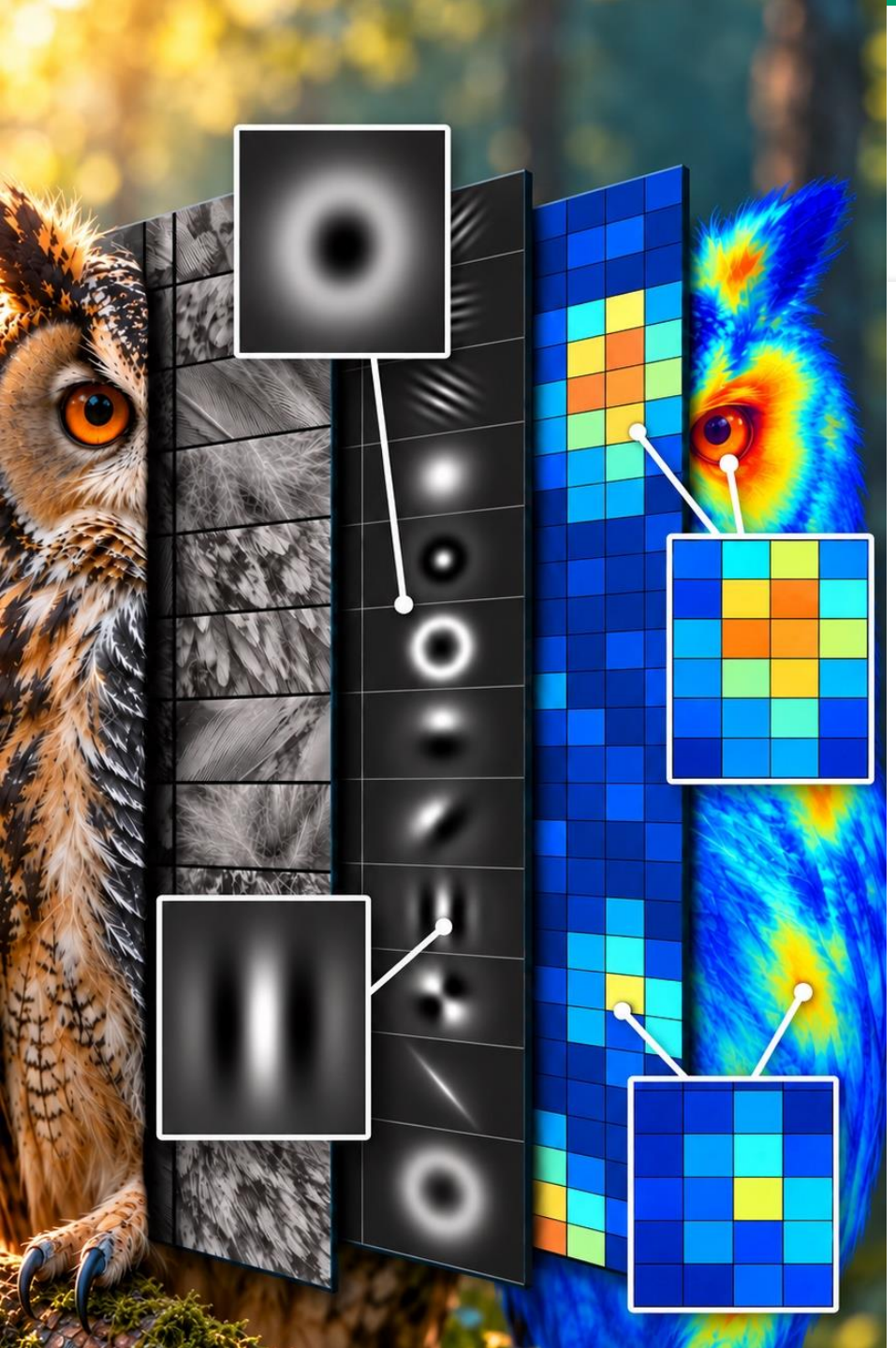


UNIVERSITÄT BAYREUTH

Applied Artificial Intelligence
02 – Initiation: Discussion & Exercise

Niklas Kühl
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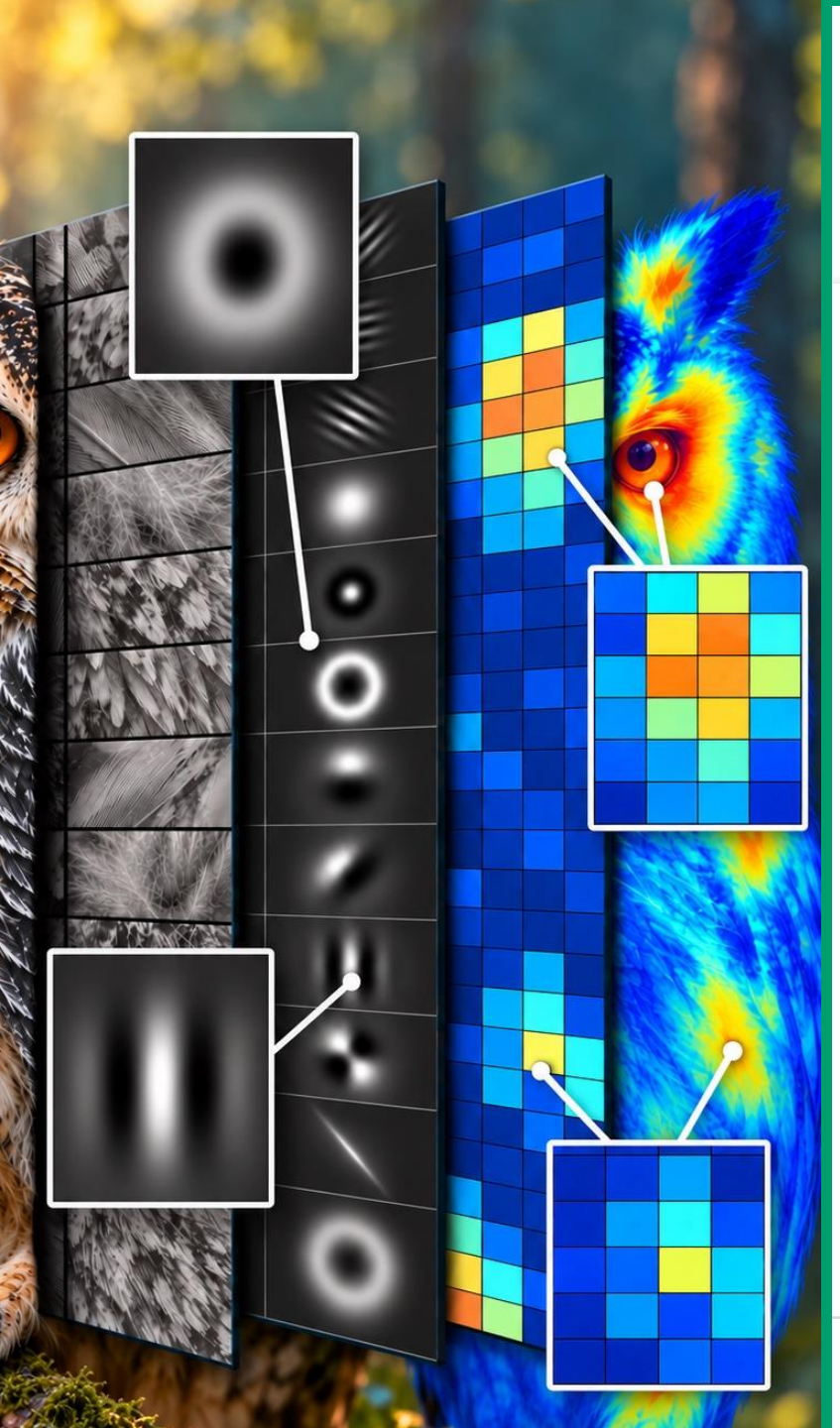
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University of Bayreuth

Lecture Schedule

	Calender Week / Lecture day	Lecture (1.5 h) – asynchronous (flipped classroom – online)	Discussion & Exercise – Tuesday, 14:15-17:45 (S 62, building RW I)
	CW 43 / 20.10.26	0: Welcome 1: Fundamentals	0: Introduction & Setup 1: Fundamentals
today	CW 44 / 27.10.26	2: AI Lifecycle: Initiation	2: Initiation
	CW 45 / 03.11.26	3: AI Lifecycle: Modeling & Evaluation	3: Modeling & Evaluation
	CW 46 / 10.11.26	4: AI Lifecycle: Deployment	4: Deployment
	CW 47 / 17.11.26	5: AI Lifecycle: Concept Drift	5: Concept Drift
	CW 48 / 24.11.26	6: AI in Systems: Foundation Models	6: Foundation Models
	CW 49 / 01.12.26	7: AI in Systems: System-Wide Learning	7: System-Wide Learning
	CW 50 / 08.12.26	8: AI in Systems: Human-AI Collaboration	8: Human-AI Collaboration ***& Project Kickoff***
	Winter break & Happy Coding :) *This means no sessions in CW 51 (15.12.26) and CW 52 (22.12.26)*		
	CW 2 / 12.01.27		***Project & Code Presentation / Defense*** *(Project & Code submission modalities will be announced in the lectures and e-learning)*
	CW 3 / 19.01.27	9: Creative AI	9: Creative AI
	CW 4 / 26.01.27	10: Ethics of AI	10: Ethics of AI + Research corner & Wrap-up
	CW 5 / 02.02.27		**Guest-Talk (TBD)**

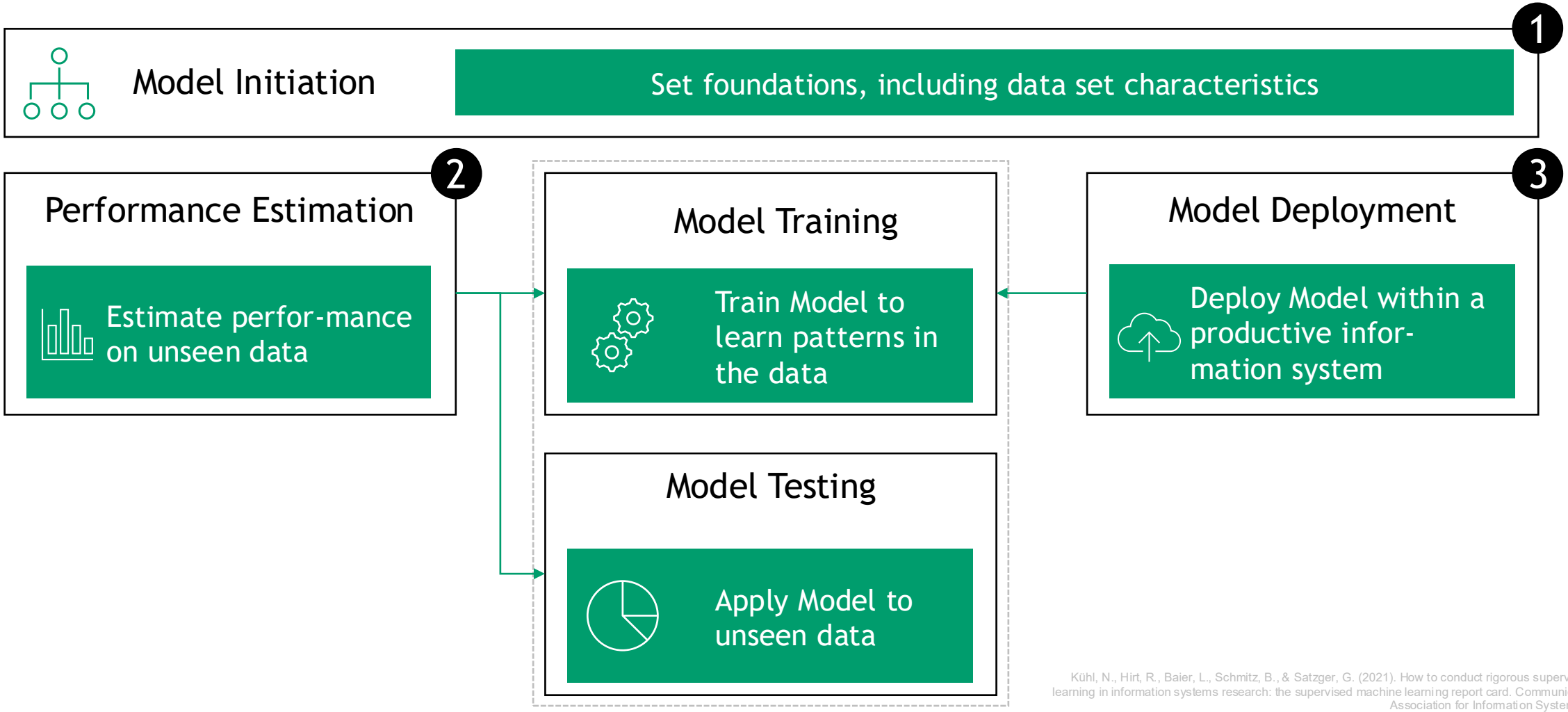
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1. Discussion
2. Exercise

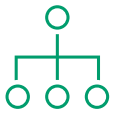
The AI Lifecycle

Required steps for supervised machine learning.



Kühl, N., Hirt, R., Baier, L., Schmitz, B., & Satzger, G. (2021). How to conduct rigorous supervised machine learning in information systems research: the supervised machine learning report card. Communications of the Association for Information Systems, 48(1), 46.

The AI Lifecycle
The initiation consists of various components.



Model Initiation

1

Set foundations, including data set characteristics

A **Problem Statement**
(e.g. binary classification, regression, etc.)

B **Data Gathering**
(e.g. rigor, labeling)

C **Sampling**
(e.g. stratified sampling from lager data pool)

D **Data Distribution**
(e.g. amount of classes, share, distribution)

E **Data Quality**
(e.g. sparsity, noise, complexity

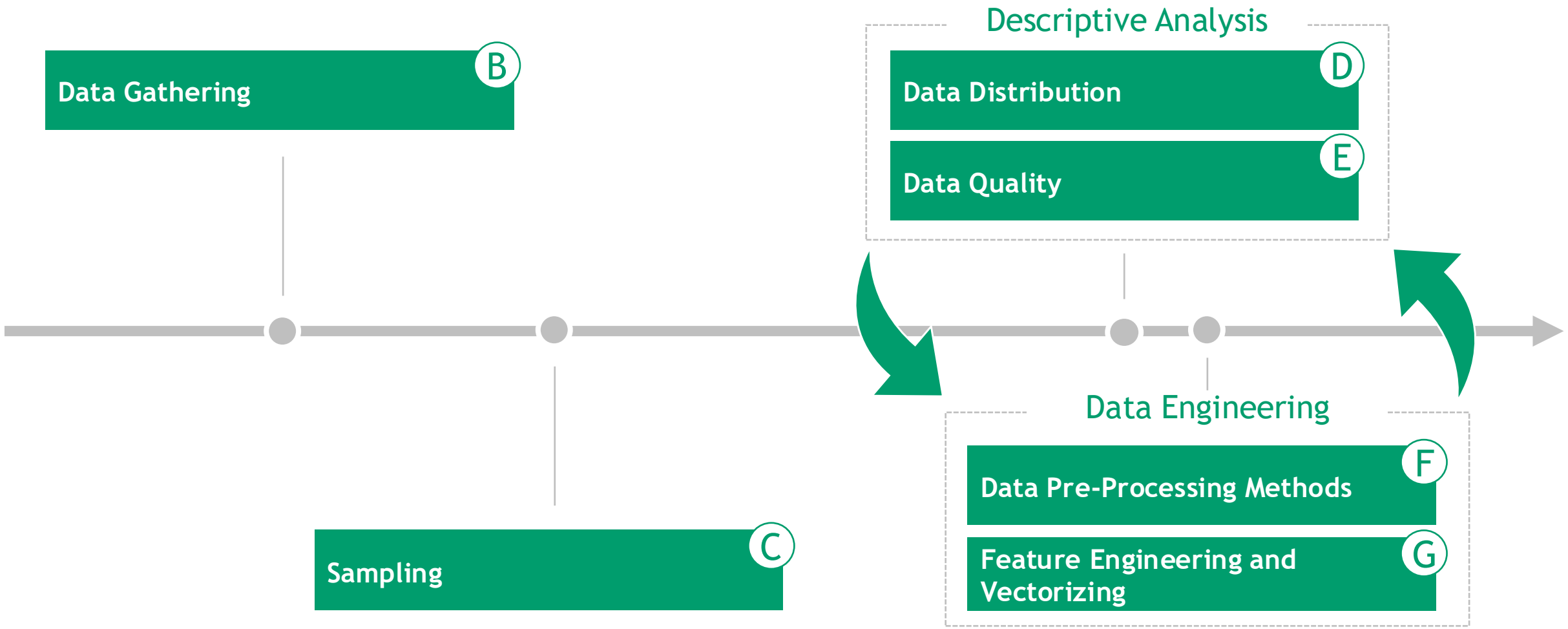
F **Data Pre-Processing Methods**
(which and why)

G **Feature Engineering and Vectorizing**
(which and why)

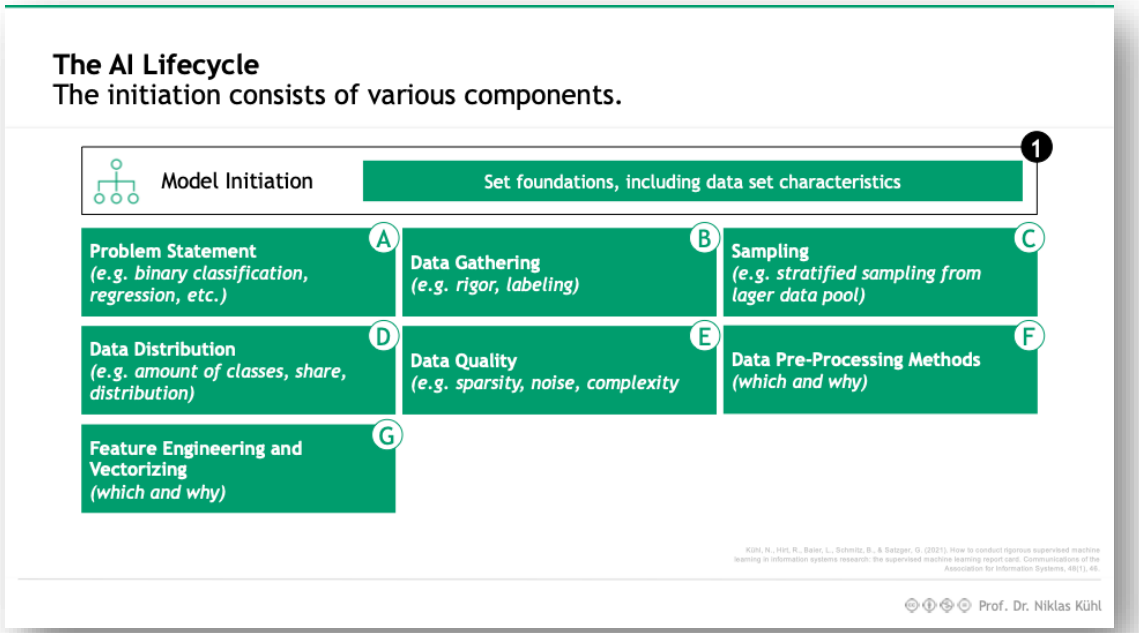
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The AI Lifecycle

Simplified process of initiation includes iterative procedure.



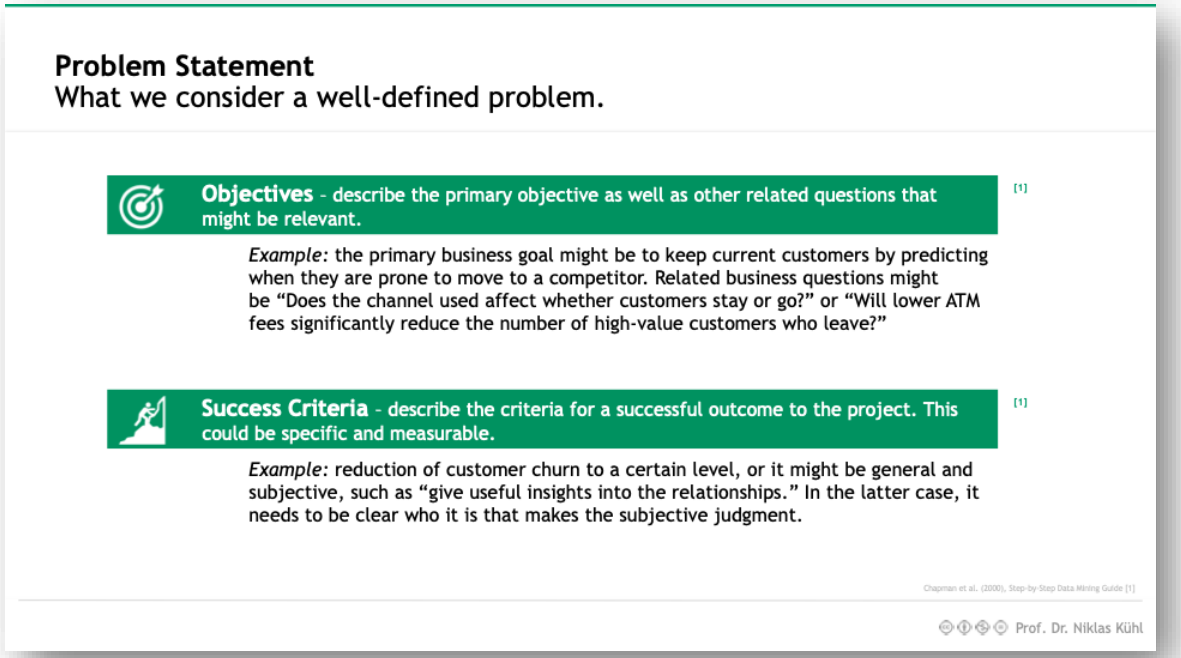
Discussion
Overall Learnings



Why is the ***combination*** of all these steps important?

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Discussion
Problem Statement Examples

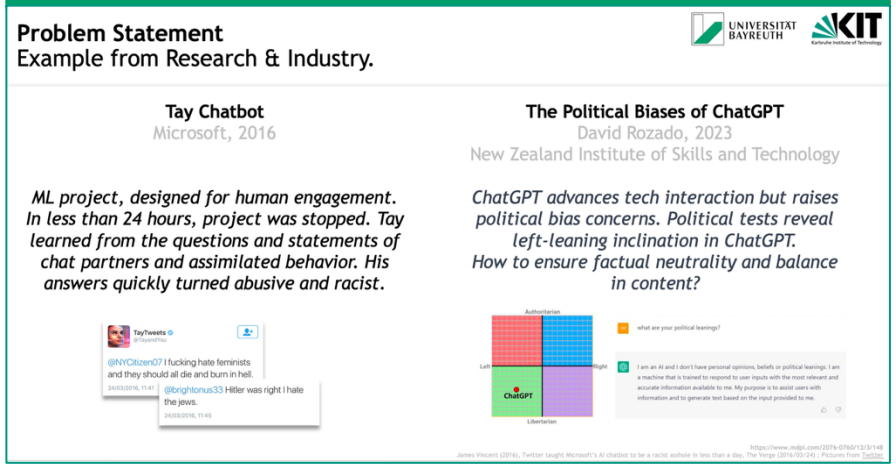


Do you have examples where the problem statement was exceptionally well understood — or the opposite?

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Discussion
Questionable model initiation.

Lecture reference



ChatBot Tay:

- What was the “problem statement” that Microsoft defined?
- Which “problem statement” should they have defined instead?
- What implications does it have on the data gathering?

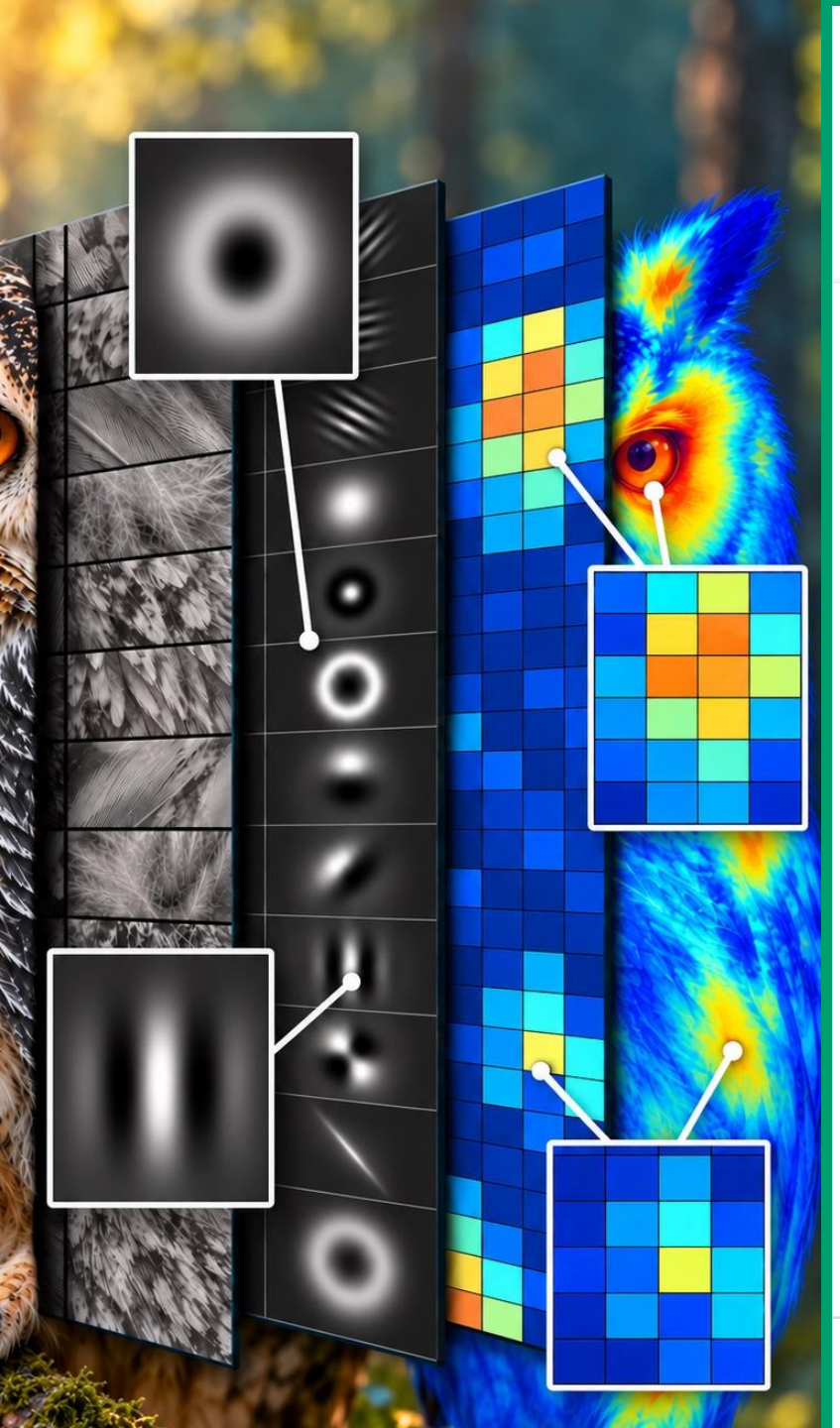
Triage in Medicine

- Which objective are we optimizing? What is a life worth?

ChatGPT

- Should a model be initiated with data that leads to left & libertarian views?
- How do we quantify political orientation? What’s the frame of reference to evaluate AI on?
- Where do you draw the line on whether a model should generate answers vs. blocking the answer and providing bin answers?

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1. Discussion
2. Exercise